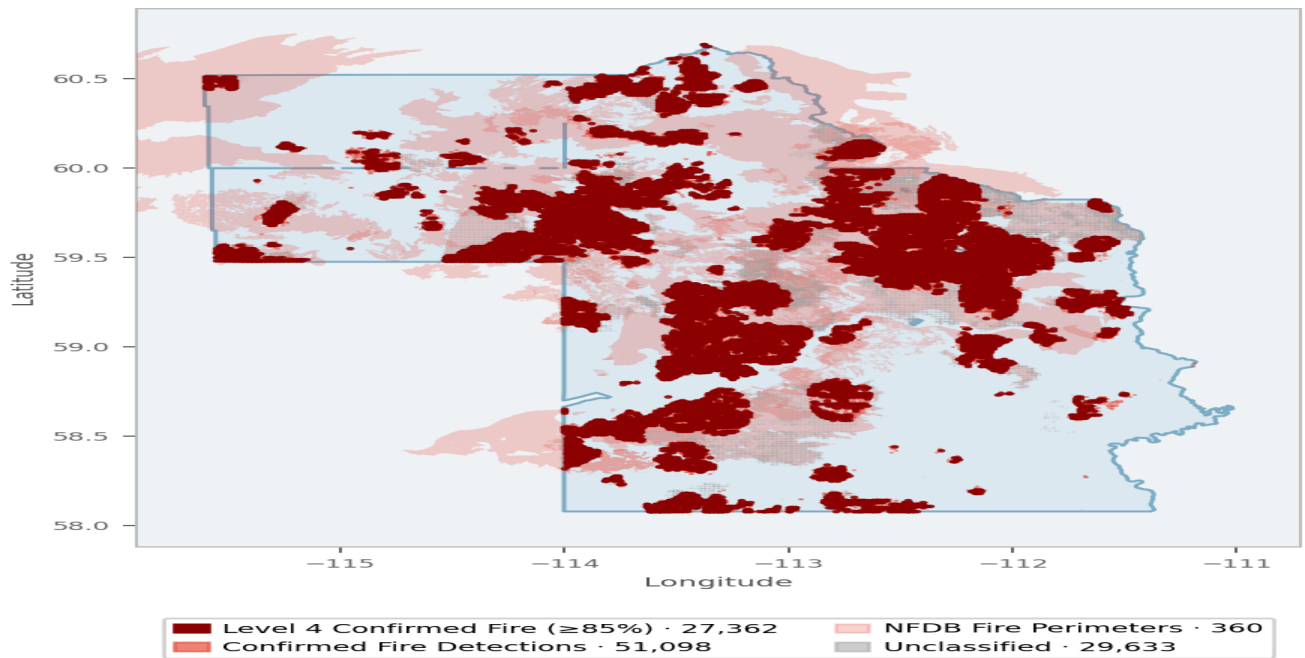


FireWatch

Wildfire Detection Validation — Wood Buffalo National Park

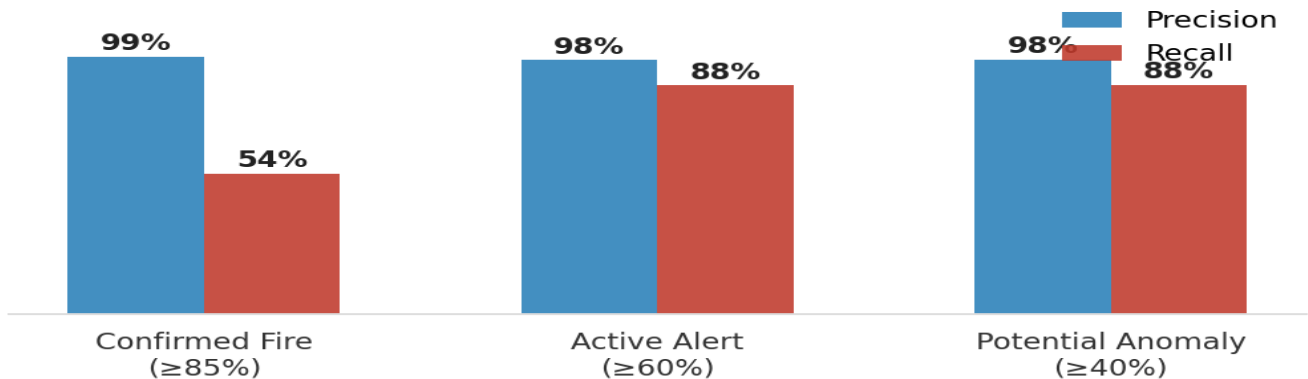
Prepared for Parks Canada & Alberta Wildfire · April 2026

98% Precision At Active Alert tier · AB/SK 2015–2023	88% Recall Of all NFDB-confirmed fires in park	51,098 Confirmed Fires Matched to ground truth
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Satellite fire detections 2015–2023 matched against NFDB confirmed fire perimeters (red shading). Dark red clusters = Level 4 Confirmed Fire (score $\geq 85\%$). Detection density closely follows confirmed NFDB perimeter locations.

Algorithm Performance by Confidence Tier



Validated against 468,000 confirmed fire clusters across AB/SK 2015–2023, including oil sands and industrial heat sources in Alberta.

FireWatch

Methodology & Pilot Program

The Problem With Raw Satellite Data

NASA's FIRMS system publishes thousands of thermal anomaly detections daily across VIIRS, MODIS, Landsat, and GOES satellites. The raw feed is extremely noisy — industrial heat sources, gas flares, and sensor artefacts are indistinguishable from real fires at the pixel level. Emergency managers cannot act on raw data alone.

How FireWatch Works

FireWatch fuses raw detections across all four satellite sources and computes a single validated confidence score per detection cluster — enabling agencies to see a ranked, prioritized alert feed instead of a wall of pixels.

Sensor	Weight	Resolution	Update Rate
VIIRS (NOAA-20 + SNPP)	0.90	375m	~12 hrs
MODIS (Terra + Aqua)	0.10	1km	~12 hrs
Landsat	0.10	30m	16 days
GOES	0.15	2km	10 min

Weights were empirically optimized via grid search across 1,540 weight combinations, evaluated against 468,000 confirmed fire clusters from the NFDB 2015–2023. A multi-sensor agreement bonus rewards clusters where independent satellites corroborate each other.

Performance Results — AB/SK Validation (2015–2023)

Confidence Tier	Threshold	Precision	Recall	F1
Confirmed Fire	≥85%	99.3%	54.2%	0.701
Active Alert	≥60%	98.1%	88.1%	0.928
Potential Anomaly	≥40%	98.1%	88.4%	0.930

Active Alert tier highlighted as primary operational tier — highest F1 score. Precision measures how often a surfaced alert is a real fire. Recall measures what proportion of all real fires are surfaced.

2026 Fire Season Pilot Program

FireWatch is offering a no-cost pilot for the 2026 fire season. Pilot partners receive near real-time validated alerts for their region — updated every 15 minutes — with four adjustable confidence tiers and area-of-interest controls. Configuration takes 48 hours.

To join — reply to this email or contact hello@outofthisworld.ca

Data: NASA FIRMS (VIIRS C2, MODIS C6.1, Landsat NRT) · CWFIS/NFDB fire perimeter polygons · Parks Canada cadastral boundary. Analysis period: May–Sep 2015–2023. Generated April 08, 2026.